Grace Y. Mahoney

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TECHNICAL SKILLS

Programming Languages: C/C++, Java, Python, MATLAB, HTML/CSS, ROS2, JavaScript, Typescript, SQL
Software Development: OOP, Data Structures & Algorithms, REST APIs, Git, Version Control
Framework & Databases: React, Node.js, Express.js, MongoDB, SQL
Tools and Platforms: Linux, Git, Render, Visual Studio Code, MATLAB, MuJoCo
Machine Learning & AI: XGBoost, Bi-LSTM, SHAP, Model Evaluation, Feature engineering,GitHub Copilot

EDUCATION

Worcester Polytechnic Institute (WPI) - Worcester, MA | GPA: 3.8/4.0
Bachelor of Science in Robotics Engineering *Aug 2023 - May 2027*
Bachelor of Science in Computer Science *Aug 2023 - May 2027*
Dean's List Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026
Relevant Course Work: Algorithms, Object-Oriented Design Concepts, Foundations of Computer Science, Software Engineering, Database Systems I, Systems Programming Concepts, Operating Systems, Data Mining and Knowledge Discovery in Databases, Machine Learning

PROJECTS

Symbiotic Multi-Agent Construction (SMAC) 7.0 | *Aug 2026 – May 2027*
- Developing a *MuJoCo-based simulation* of 5-DOF Inchworm robots and intelligent Smart Blocks with kinematic constraints, collision detection, and asynchronous communication.
- Developing *decentralized swarm algorithms* using stigmergic cell-weight gradients to guide navigation, material retrieval, collision avoidance, and autonomous construction from locally available information.

Autonomous TurtleBot Navigation with ROS2 | *Oct 2025 – Dec 2025*
- Implemented *A*-based path planning* with *LiDAR* obstacle detection and *AMCL* localization for autonomous navigation.
- Solved the "kidnapping problem," enabling the robot to re-localize and autonomously return to its home position using *ROS2 navigation* stack.

PhotoBucket Collaborative Web Application | *Aug 2025 – Oct 2025*
- Developed a collaborative bucket-list web app in a team of 3 using *TypeScript, React, Node.js, Express,* and *MongoDB*; deployed via *Render* with secure authentication and persistent session management.
- Implemented *real-time collaboration*, visual completion tracking with photo uploads, and a responsive UI with *TailwindCSS* and *AceternityUI* for interactive goal management and memory documentation.

Smart Parking Availability with Real-time Knowledge & AI | *May 2025 – July 202*
- Designed a *smart parking prediction system* using LADOT and *NOAA* data, achieving *99% F1-score* with XGBoost by modeling *temporal, spatial,* and *weather-based features.*
- Compared *ensemble* and *sequential models* (Bagging, XGBoost, Bi-LSTM) with *SMOTE balancing* and *SHAP interpretability*; proposed real-time deployment on *edge devices.*

WORK EXPERIENCE

Dog Walker / Pet Sitter, Rover and Independent | *2016 - Present*
- Demonstrated reliability, time management, and accountability while managing recurring client schedules and responsibilities.

Suite Runner, Polar Park*, Worcester, MA* | *May 2026 – Present*
- Demonstrated organization, responsiveness, and teamwork while preparing event suites, coordinating catered food and supplies, supporting servers, and completing post-event cleanup.

ACTIVITIES

Treasurer, Autonomous Underwater Vehicle Team (AUV)
Public Relations Committee, Society of Asian Scientists and Engineers (SASE)